

Two Shared Tasks - Membership Inference Attacks on LLMs and VLMs

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Abstract

This paper presents our work on two shared tasks in membership inference attacks (MIA). First is detecting whether clinical discharge notes were used to fine-tune an autoregressive language model, and second was whether image-caption pairs were used to fine-tune a vision-language model. For the LLM task, we implement and compare four methods including perplexity scoring, Min-K%, neighbourhood comparison, and a self-prompt calibration ensemble which achieved a best Kaggle AUC of 0.749. For the VLM task, we developed an iterative white-box model using XGBoost with features drawn from loss, gradient norms, attention transitions, image corruption sensitivity, and cross-modal embedding similarity, and that achieved a best Kaggle AUC of 0.837. One of the key findings was that a structural pattern provided a stronger membership signal than any individual attack method, which highlights the importance of modelling with data analysis in MIA tasks.

1 Introduction

Membership Inference Attack (MIA) refers to the ability to determine which specific data points were used to train a machine learning model. Given a trained model and a candidate sample, an attacker attempts to infer whether that sample was part of the training set. In cases involving clinical records or copyrighted material, there is a significant privacy risk and potential legal consequences if confirmed.

Modern artificial intelligence (AI) systems, including large-language models (LLMs) and vision-language models (VLMs), are increasingly fine-tuned on sensitive private data, and there is a growing concern around this. MIA has two major purposes, and while it can pose a direct privacy threat, it can also be used by security researchers and auditors to verify whether training data is legally sourced and to diagnose the degree

to which a model will memorise versus generalise (Hengyu Wu, 2025).

Most prior MIA work focuses on output-level signals such as loss and perplexity (Jawad et al., 2026). These signals are intuitive, but in our experiments they were often too noisy to be useful at low false-positive rates, which is where real-world privacy auditing operates (Zaree et al., 2026). VLMs present an additional layer of difficulty because membership is a joint property of both visual and linguistic components, and therefore remains comparatively underexplored (Hu et al., 2025; Li et al., 2024b).

This paper investigates fine-tuned LLMs (SmolLM2-135M and SmolLM2-360M) on clinical discharge notes, and a fine-tuned VLM (SmolVLM-500M) on image-caption pairs. Our main contributions were implementing and comparing multiple LLM attack methods including perplexity scoring, Min-K%, neighbourhood comparison, and a self-prompt calibration ensemble. The best Kaggle AUC achieved with this model was 0.749. We also develop an iterative white-box ensemble for VLM MIA incorporating reference loss delta, attention transition features, image corruption sensitivity, Min-K%+, and cross-modal embedding similarity, achieving a best Kaggle AUC of 0.837. Lastly, we identified a structural pattern in sample IDs that acts as a strong membership signal in the VLM task.

2 Related Works

The most foundational approach to MIA is loss-based scoring, which is where a sample is classified as a member if its loss falls below a threshold derived from the model’s training loss distribution. This approach is intuitive and easy to implement, however it is limited if used alone and is also not always reliable (Hengyu Wu, 2025). The neighbourhood comparison method, which evaluates isolated

samples, generates several deviations by replacing nouns with similar alternatives in terms of semantics. If the original sample has a significantly lower loss than its neighbours, this is taken as a membership signal. Min-K% (?) focuses on the bottom k% of tokens by log-probability. The idea is that a model that has memorised a text will assign higher probability even to its hardest tokens, whereas a non-member will show uniformly low probabilities on those tokens. Min-K%++ takes it a step further by normalising each token’s log-probability against the mean log-probability, which makes the signal better for tokens that are rare. Wenjie Fu (2024) proposed self-prompt calibration which uses the target LLM’s own generations as a reference dataset to fine-tune a calibration model. This also addresses the memorisation signal more directly and improves AUC to approximately 0.93.

With white-box access, attacks can also observe internal model behaviour. Jawad et al. (2026), in their paper, suggested HT-MIA which focuses on the tokens that the model finds hardest to predict. The main idea is that a fine-tuned model should show the most improvement over its base version at domain-specific tokens if the text was in the training set. They test on medical and general datasets and consistently outperform baseline methods. Li et al. (2024a) show that including the top $\tau\%$ of neuron activation values as a feature improves MIA performance beyond what output probabilities alone can achieve. They use a shadow model trained on data from the same distribution as the target model to train a classifier that combines multiple feature types. Zaree et al. (2026) take this further with AttenMIA which is the first attention-based white-box MIA framework for LLMs. Rather than relying on output statistics, it extracts features from how attention distributions shift across consecutive layers. This is extremely effective in the low-false-positive regime and achieves up to 0.996 AUC on WikiMIA with LLaMA-2-13B.

MIA for vision-language models is a newer and less explored area. Hu et al. (2025) find that membership signal is weak at the individual example level but becomes detectable when classifying sets of examples together. They also showed that changing generation temperature results in differences in how the model reproduces text for member versus non-member images. Li et al. (2024b) introduce VL-MIA which is a benchmark designed specifically for VLM membership inference which ex-

tends token-level scoring to the cross-modal setting by using Rényi entropy. On the black-box side, Yin et al. (2025) proposes KCMP which probes visual memorisation by masking objects in an image and asking the model to identify them from alternatives. Wu et al. (2026) propose ICI MIA which is based on the observation that VLMs are more sensitive to image corruption for member images than non-member images. Bian et al. (2026) take a different angle with PIPRA, and show that the prompt vector itself leaks membership information due to its embedding space, enabling output-free membership inference. Gomez (2026) highlights an the problem of how many reported results are inflated because non-member sets are drawn from different distributions. Their work shows that black-box attacks do not work well when this distributional cue is removed, and that white-box loss-gradient attacks are significantly more reliable.

3 Data

3.1 LLM

The data used for the LLM task is accessible through the UBC SLIME Hugging Face account and consists of patient discharge notes. Each sample has a numerical ID, a text field containing the discharge note, and an is_member label. The label is 0 if the document was not seen during model training and 1 if it was. The task is therefore a binary classification where, given a candidate document and a fine-tuned model, determine whether the document was part of the training set.

The discharge notes are structured clinical documents that include personal information with sensitive details redacted, hospital course, diagnosis, treatment, complications, and follow-up instructions. The dataset is split into train, validation, and test sets. The train and validation sets include the is_member label, while the test set does not. The summary statistics are shown in table below.

Split	Samples	Mean Length	Max Length	Min Length
Train	50,000	1956.1	5571	695
Validation	10,000	1949.7	4818	761
Test	15,000	1956.9	5312	716

3.2 VLM

The data used for the VLM task is also accessible through the UBC SLIME Hugging Face account and has multiple image-caption pairs. Each sample has an ID label, an image, and a caption

stored as plain text. Each sample belongs to one of four types depending on what the target model saw during training: `seen_both`, `unseen_both`, `seen_img_unseen_txt`, and `unseen_img_seen_txt`. The task is to identify only the `seen_both` samples which makes the task harder than a straightforward binary split because the model has to figure out the full membership from partial exposure. The summary statistics are shown in the table below.

Split	Samples	Mean Length	Max Length	Min Length
Train	6,000	229.5	776	69
Validation	1,200	224.4	698	77
Test	6,000	228.8	1060	62

4 Methods

We used perplexity scoring for our baseline in the LLM task. We take each sample and forward it through the model to compute its loss, which is then converted to perplexity. A lower perplexity suggests that the model has seen it before and therefore can predict it, which is weak evidence of membership. The validation set was then used to determine an optimal threshold via an ROC curve and Youden’s J statistic, which identifies the threshold that balances sensitivity and specificity. Samples below the threshold are classified as members. This model was run on a Kaggle Notebook with P100 GPU acceleration and achieved 0.56 accuracy.

For the VLM task, the baseline model takes the loss on the assistant text tokens only, excluding the user prompt tokens. The idea is that all prompt text samples are similar so there is very little discriminative signal. The best threshold is again determined using $\text{TPR@FPR}=0.1$ on the validation set. We had to use GPU for this task because CPU was too slow.

4.1 LLM Methods

For the LLM task, we implement four attack methods. First, we have `Min-K%`, which scores each sample by averaging the log-probabilities of the bottom $k\%$ of tokens by probability rank. If a model has memorised a text, it will assign higher probability even to harder tokens. Each sample requires only a single forward pass. The per-token log-probabilities were noted and the average of the bottom $k\%$ was taken. The best result was at $k = 30\%$ which achieved an AUC of 0.670 and $\text{TPR@FPR}=0.1$ of 0.0122 on validation.

Next, the Neighbourhood Comparison method generates many versions of each sample by iden-

tifying nouns and masking them at a ratio of 0.15. Then, a BERT model substitutes each masked token with an appropriate alternative. We calculate the membership score by looking at the difference between the original sample’s loss and the average loss across five altered versions. A large positive difference means the model has specifically memorised the original. Due to the high computational cost, the threshold was estimated on only the first 200 validation samples and the best result achieved was approximately 0.50 AUC.

The Self-Prompt Calibration Ensemble gives the model a part of the text and measures how confidently it predicts the rest. With familiar text, the model is expected to have higher confidence. Multiple signals are extracted from a single forward pass and combined using logistic regression with L2 regularisation, trained on 30,000 labelled examples. This achieved 0.75 AUC which was the best result on the LLM task. Lastly, the Multi-Feature Ensemble expands on the above methods by adding a loss ratio between the fine-tuned and base SmolLM model, and the top $\tau\%$ neuron activations from the last hidden layer selected using a Random Forest. If the fine-tuned model has significantly lower loss than the base model on a sample, this is a strong indicator of membership.

4.2 VLM Methods

For the VLM task, we developed an iterative ensemble across six models. The foundation is a multi-feature XGBoost classifier trained on all 6000 training samples. The initial feature set looks at basic loss, loss variance, perplexity, zlib ratio, Min-K score ($k = 20$), logit gap, gradient norm, layer similarities, image sensitivity, and others, achieving approximately 0.61 AUC. The second iteration adds three new features (1) a loss delta between the base and fine-tuned SmolVLM-500M-Instruct model, (2) gradient norm of the `lm_head` parameters, and (3) temperature similarity which is measured via ROUGE-2 F1 at $T \in \{0.2, 1.0\}$. This brings the total feature count to 20. The next iteration adds attention transition features inspired by Zaree et al. (2026). The inter-layer attention consistency is measured using cosine similarity between consecutive layer attention distributions. The second feature also captures how attention patterns shift under word-level perturbations.

Next we identified the structural patterns in the sample IDs. IDs beginning with `mix_SI_UT` or `mix_UI_ST` consistently correspond to non-

members, while IDs beginning with cauldron_In correspond to either seen_both or unseen_both. This narrows the classification problem from four categories to two and produced about a 20% increase in AUC, with a final score of 0.841. Min-K%++ was also introduced at this stage.

The fifth version adds image corruption sensitivity which was inspired by Wu et al. (2026). There is a slight corruption that is applied to the input image and the loss on the corrupted version is computed. Member images are expected to have a larger loss increase if there is corruption because the model has overfit to their exact pixel distributions.

The last version had cross-modal embedding similarity where the last hidden layer is mean-pooled over visual and text token positions separately and therefore produces two L2-normalised vectors. The cosine similarity between these vectors is used as a feature, as well as similarity deltas relative to random distractors. This model achieved a validation AUC of 0.8435 and a final Kaggle AUC score of 0.8351.

4.3 Generative AI Usage

This report uses Generative AI to format written answers into LaTeX appropriate code and create tables. Throughout the coding process, we also used Generative AI to iterate through model approaches and solve issues we found. A common usage was explaining a feature that we wanted added to our current list of features and then adding the updated code.

5 Experiments

All the LLM experiments were run on a Kaggle Notebook with P100 GPU acceleration or T4 x2 GPU acceleration. The target models for the LLM task are UBC-SLIME/colx_531_smol1m2-135m and UBC-SLIME/colx_531_smol1m2-360m. All the inputs were tokenized with a max sequence length of 512 tokens. The VLM experiments were also run on Kaggle Notebooks with GPU acceleration, with some run on GPU via Google Colab and the target model was UBC-SLIME/colx_585_vlm, fine-tuned from SmolVLM-500M-Instruct, with image resolution capped at 512 pixels on the longest edge. The base model used for loss delta comparisons is HuggingFaceTB/SmolVLM-256M-Instruct. All models were evaluated using AUC as the primary metric and TPR@FPR=0.1 as the secondary one.

The predictions were then submitted to the Kaggle leaderboard, which evaluates on approximately 50% of the test data.

The LLM baseline used SmolLM2-135M with a maximum token length of 512. The perplexity was calculated with $e^{\mathcal{L}}$ where $\mathcal{L}$ is the mean cross-entropy loss, and the optimal threshold of 5.044 was found using Youden’s J statistic on the validation set. For Min-K%, it only needed a single forward pass per sample. We tried $k \in \{10, 20, 30\}\%$ and $k = 30\%$ worked best on validation. The neighbourhood comparison was the most difficult method to run. The nouns were masked at a 0.15 ratio and filled in using bert-base-uncased, producing five altered versions per sample. Because of the runtime, we could only calibrate the threshold on the first 200 validation samples. The score is the difference between the original loss and the mean loss across the five alternatives. Lastly, the self-prompt calibration ensemble used SmolLM2-360M with a prefix ratio of 0.3 and batch size of 32. Signals were combined using a GradientBoostingClassifier trained on 15,000 examples.

For the VLM baseline, prompt tokens were masked with -100 so that loss would then only be computed on the assistant response. The threshold was set using TPR@FPR=0.1 on the validation set. Next, the XGBoost ensemble used 1,000 estimators, depth 3, learning rate 0.01, subsampling 0.7, $\alpha = 0.1$, $\lambda = 1.0$, with early stopping at 30 rounds. All features were scaled with StandardScaler. Text variations averaged 5 word swaps per sample, temperature similarity was measured with ROUGE-2 F1 at $T \in \{0.2, 1.0\}$, and the distractor pool for image sensitivity had 10 images.

6 Results

6.1 LLM Task

Table 1 shows the results for the LLM task. The self-prompt calibration ensemble was the strongest LLM method at 0.749 AUC. Overall, the clinical discharge notes were structured and repetitive which made it harder for loss-based signals to distinguish members from non-members. The top team on the leaderboard achieved 0.955 AUC, which suggests there is significant room for improvement that we did not reach.

Method	AUC	TPR@FPR=0.1
Perplexity Baseline	0.560	–
Neighbourhood Comparison	0.50	–
Min-K% ($k = 30\%$)	0.670	0.0122
Self-Prompt Calibration	0.749	–
Multi-Feature Ensemble	0.749	–

Table 1: LLM task results. AUC scores are from the Kaggle private leaderboard or otherwise from validation.

6.2 VLM Task

Table 2 shows the results for the VLM task. The loss baseline sat at approximately 0.50, showing that assistant text loss alone is not a useful signal for this task. The initial XGBoost ensemble with 26 features brought this to 0.61, and subsequent iterations adding loss delta, gradient norm, temperature similarity, and attention features produced modest improvements.

The biggest jump came from the ID-prefix observation. Once the `mix_` prefixed samples were assigned directly as non-members and the classifier only needed to distinguish between `seen_both` and `unseen_both` within the `cauldron_In` subset, the AUC jumped to 0.841. This single observation was worth more than all the modelling iterations that came before it. The image corruption and embedding similarity features in the final two models maintained this level of performance, ending at a final Kaggle AUC of 0.837 and placing 3rd out of 8 teams.

The TPR@FPR=0.1 of 0.3633 on the final model shows that even at a strict false positive constraint, the model still identifies a meaningful proportion of true members. This is significantly better than the LLM task, where the Min-K% TPR@FPR=0.1 of 0.0122 shows the attack barely functions at low false positive rates.

Method	AUC	TPR@FPR=0.1
Loss Baseline	~0.50	–
XGBoost Ensemble (26 features)	0.614	–
+ Loss Delta, Grad Norm, Temp Sim	~0.61	–
+ AttenMIA Features	~0.61	–
+ ID-Prefix Signal, Min-K%++	0.841	–
+ Image Corruption (ICIMIA)	~0.837	–
+ Embedding Similarity	0.837	0.3633

Table 2: VLM task results. AUC scores are from the Kaggle public leaderboard where available, otherwise from validation.

7 Conclusion

This paper explored membership inference attacks in fine-tuned LLMs on clinical discharge notes and a fine-tuned VLM on image-caption pairs. In both tasks, simple output-level signals like loss and perplexity were weak on their own, and meaningful performance required either a supervised ensemble combining multiple signals and a deeper understanding of the data.

On the LLM task, the best result of 0.749 AUC came from the self-prompt calibration ensemble. The clinical documents, due to their structured and repetitive organisation, made it hard to distinguish which individual samples stood out. On the VLM task, the most important takeaway was that understanding the data matters significantly. The ID-prefix discovery produced a larger AUC improvement than all the other modelling iterations and the final ensemble had a score of 0.837 AUC.

The main limitations were compute and collaboration. Longer runtimes meant fewer iterations, and there needed to be a more efficient way to iterate on each others’ models rather than working on it individually. A cleaner shared pipeline and a better set up of our GitHub repository from the start would have changed how we worked. We also see scope in exploring alternative metrics.

8 Further Scope

There are a few directions worth pursuing with more time, tokens and compute. For the LLM task, using both SmolLM2-135M and SmolLM2-360M together as a reference pair could potentially strengthen the membership signal. For the VLM task, the most promising direction is building a stronger classifier specifically on the `cauldron_In` subset, since that is where the remaining uncertainty lives after the ID-prefix signal is applied. Also, using a larger distractor pool for image sensitivity features and experimenting with more corruption types for the ICIMIA feature are also worth exploring.

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